# Counterexamples to $\xi$-reduction and minimal-integral duality below one

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## Abstract

We construct counterexamples on the unit disk to the $\xi$-reduction formula and minimal-integral duality for every $0 < p < 1$, using higher derivative orders and three-term polynomial perturbations. For $p \geq 1$, we prove minimal-integral duality on arbitrary pseudoconvex domains. Thus $p \geq 1$ is the optimal universal exponent range for both identities.

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## 1 Introduction

For $p > 0$ and a homogeneous polynomial $H$ of degree $k$, the higher order $p$-Bergman kernel $K_D^{H,p}(z)$ and, for a functional $\xi \in \ell_1^{(n)}$, the $p$-Bergman kernel with respect to $\xi$ $K_{\xi,D,p}(z)$ are defined in Section 2. Bao–Guan–Sun [3] proved that for every $p \in [1, +\infty)$,

$$K_D^{H,p}(z) = \inf_{\xi \in S_H} K_{\xi,D,p}(z),$$

where $S_H$ is the affine family of functionals whose degree-$k$ part is prescribed by $H$; see Theorem 1.7 in [3].

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We show that the range $p \geq 1$ is optimal for a theorem valid for all orders: for every $0 < p < 1$, a counterexample exists on the unweighted unit disk, with the order chosen sufficiently large.

THEOREM 1.1. *Let $0 < p < 1$, and choose an integer $k \geq 1$ such that*

$$(1-p)(kp+2)^2 > (2-p)^2. \tag{1.1}$$

*For $D = \mathbb{D} \subset \mathbb{C}$, $H_k(z) = z^k/k!$, and the base point $0$,*

$$\inf_{\xi \in S_{H_k}} K_{\xi,\mathbb{D},p}(0) > K_{\mathbb{D}}^{H_k,p}(0) = \frac{kp+2}{2\pi}. \tag{1.2}$$

*For every $p \in (0,1)$, condition (1.1) holds for all sufficiently large integers $k$. In particular, the $\xi$-reduction formula fails for some higher order kernel on $\mathbb{D}$ for every $0 < p < 1$.*

For $k = 1$, condition (1.1) is equivalent to $0 < p < 2(\sqrt{2}-1)$ and includes the first-order example $p = 1/2$. Increasing $k$ produces counterexamples for exponents arbitrarily close to 1. The order is allowed to depend on $p$; no single fixed order is asserted to satisfy (1.1) for every $p < 1$. As shown in Theorem 3.1, the same construction also gives $B_p(F, I, \mathbb{D}) < C_{F,I,p}(\mathbb{D})$ for every $0 < p < 1$.

# 2 The $\xi$-reduction fails for $0 < p < 1$

## 2.1 Definitions and normalization

Let $D \subset \mathbb{C}^n$ be a domain, $p > 0$, and let $H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$ be homogeneous of degree $k$. Write $P_H(f) = \sum_{|\alpha|=k} a_\alpha D^\alpha f$ and $A^p(D) = \mathcal{O}(D) \cap L^p(D)$, with respect to Lebesgue measure. Following [3], put

$$\ell_1^{(n)} := \Big\{ \xi = (\xi_\alpha)_{\alpha \in \mathbb{N}^n} : \sum_{\alpha \in \mathbb{N}^n} |\xi_\alpha| \rho^{|\alpha|} < +\infty \text{ for every } \rho > 0 \Big\}.$$

For $\xi \in \ell_1^{(n)}$ and a holomorphic germ $f$, the pairing is

$$(\xi \cdot f)(z) = \sum_{\alpha \in \mathbb{N}^n} \xi_\alpha \frac{D^\alpha f(z)}{\alpha!}.$$

The affine family $S_H$ consists of the functionals satisfying

$$\xi_\alpha = a_\alpha \alpha! \quad (|\alpha| = k), \qquad \xi_\alpha = 0 \quad (|\alpha| > k),$$

with the entries of degree less than $k$ free. The kernels are

$$K_D^{H,p}(z) = \sup\Big\{ |P_H(f)(z)|^p : f \in A^p(D),\ f^{(j)}(z) = 0\ (0 \leq j < k),\ \|f\|_{L^p(D)} \leq 1 \Big\},$$

$$K_{\xi,D,p}(z) = \sup_{f \in A^p(D) \setminus \{0\}} \frac{|(\xi \cdot f)(z)|^p}{\|f\|_{L^p(D)}^p}.$$

For a nonzero functional, the latter is equivalently

$$K_{\xi,D,p}(z) = \left( \inf_{\substack{f \in A^p(D) \\ (\xi \cdot f)(z) = 1}} \|f\|_{L^p(D)}^p \right)^{-1}.$$

On $\mathbb{D}$, write $f_j = f^{(j)}(0)/j!$ and choose $H_k(z) = z^k/k!$. Then $P_{H_k}(f)(0) = f_k$, and every $\xi \in S_{H_k}$ is of the form

$$\xi = (c_0, \dots, c_{k-1}, 1, 0, \dots), \qquad (\xi \cdot f)(0) = f_k + \sum_{j=0}^{k-1} c_j f_j. \tag{2.3}$$

The factorial normalization is only for convenience. Replacing $H_k$ by $z^k$ multiplies both sides of (1.2) by $(k!)^p$.

LEMMA 2.1. *For every $p > 0$ and every integer $k \geq 1$,*

$$\inf_{\substack{f \in \mathcal{O}(\mathbb{D}) \\ f_0 = \cdots = f_{k-1} = 0,\ f_k = 1}} \int_{\mathbb{D}} |f|^p \, dA = \frac{2\pi}{kp+2}, \qquad K_{\mathbb{D}}^{H_k,p}(0) = \frac{kp+2}{2\pi}. \tag{2.4}$$

**Proof.** An admissible function factors as $f(z) = z^k g(z)$ with $g(0) = 1$. Subharmonicity of $|g|^p$ gives

$$\frac{1}{2\pi} \int_0^{2\pi} |g(re^{i\theta})|^p \, d\theta \geq 1.$$

It follows that

$$\int_{\mathbb{D}} |f|^p \, dA \geq 2\pi \int_0^1 r^{kp+1} \, dr = \frac{2\pi}{kp+2}.$$

Equality is attained by $f = z^k$. The reciprocal identity gives the kernel. ∎

## 2.2 A three-term perturbation and its second variation

The following calculation supplies the counterexamples. Its remainder estimate is included because the order of the zero at 0 changes under the perturbation, and $|f|^p$ is not twice continuously differentiable at a zero when $p < 2$.

LEMMA 2.2. *Let $p > 0$, $k \geq 1$, and $\lambda > 0$. For real $\varepsilon$, set*

$$F_\varepsilon(z) = z^k + \varepsilon\big(z^{k-1} + \lambda z^{k+1}\big), \qquad \mathcal{J}_{k,p}(\varepsilon) = \frac{1}{\pi} \int_{\mathbb{D}} |F_\varepsilon|^p \, dA. \tag{2.5}$$

*As $\varepsilon \to 0$,*

$$\mathcal{J}_{k,p}(\varepsilon) = \frac{2}{kp+2} + \frac{\varepsilon^2}{2} \mathcal{Q}_{k,p}(\lambda) + o(\varepsilon^2), \tag{2.6}$$

*where*

$$\mathcal{Q}_{k,p}(\lambda) = \frac{p}{k} + \frac{p^2\lambda^2}{kp+4} + \frac{2p(p-2)\lambda}{kp+2}. \tag{2.7}$$

**Proof.** For $z \neq 0$, write

$$F_\varepsilon(z) = z^k(1 + \varepsilon v(z)), \qquad v(z) = z^{-1} + \lambda z.$$

The real Taylor expansion at $w = 0$ is

$$|1+w|^p = 1 + p\Re w + \frac{p^2}{4}|w|^2 + \frac{p(p-2)}{4}\Re(w^2) + O(|w|^3), \qquad |w| \leq \frac{1}{2}. \tag{2.8}$$

The remainder in (2.8) is bounded by $C_p|w|^3$ uniformly on $|w| \le 1/2$: the function $|1+w|^p$ is smooth on a neighborhood of this closed disk. We justify integrating its terms to second order. Throughout the proof, $k, p, \lambda$ are fixed; all constants may depend on these parameters, but not on $\varepsilon$ or the polar angle. Put $\delta = |\varepsilon|$ and fix $L > 2(1+\lambda)$. Take $0 < \delta < \min\{1, 1/L\}$ so that the small disk lies in $\mathbb{D}$. For $|z| \le L\delta$, the factorization

$$F_\varepsilon(z) = z^{k-1}\big(z + \varepsilon + \varepsilon\lambda z^2\big)$$

implies $|F_\varepsilon(z)|^p \le C\delta^p |z|^{(k-1)p}$. Consequently,

$$\begin{aligned}\int_{|z|\le L\delta} |F_\varepsilon|^p \, dA &\le C\delta^p \int_0^{L\delta} r^{(k-1)p+1}\, dr \\ &= O(\delta^{kp+2}) = o(\delta^2).\end{aligned} \tag{2.9}$$

The integrals over this small disk of the constant, linear, and quadratic terms in $|z|^{kp}$ times (2.8) are also $O(\delta^{kp+2})$. Indeed, their absolute values are bounded by constant multiples of

$$|z|^{kp}, \qquad \delta|z|^{kp-1}, \qquad \delta^2|z|^{kp-2},$$

respectively; all three are integrable since $kp > 0$.

On $L\delta < |z| < 1$, one has $|\varepsilon v(z)| \le \delta(1+\lambda)/|z| < 1/2$. The integral of the remainder in (2.8), multiplied by $|z|^{kp}$, is bounded by

$$C\delta^3 \int_{L\delta}^1 r^{kp-2}\, dr = \begin{cases} O(\delta^{kp+2}), & 0 < kp < 1, \\ O(\delta^3 \log(1/\delta)), & kp = 1, \\ O(\delta^3), & kp > 1. \end{cases}$$

Each expression is $o(\delta^2)$. Together with (2.9), this proves that the full-disk integral is given by the integrated Taylor polynomial with an $o(\varepsilon^2)$ error. Thus no formal differentiation at the moving zero is needed.

To compute the coefficients, for $z = re^{i\theta}$ use

$$\frac{1}{2\pi}\int_0^{2\pi} \Re v \, d\theta = 0, \qquad \frac{1}{2\pi}\int_0^{2\pi} |v|^2 \, d\theta = r^{-2} + \lambda^2 r^2, \qquad \frac{1}{2\pi}\int_0^{2\pi} \Re(v^2)\, d\theta = 2\lambda.$$

After angular averaging and radial integration against $2r\, dr$, the coefficient of $\varepsilon^2$ is

$$\frac{p^2}{2kp} + \frac{p^2\lambda^2}{2(kp+4)} + \frac{p(p-2)\lambda}{kp+2} = \frac{1}{2}\mathcal{Q}_{k,p}(\lambda).$$

The constant term is $2/(kp+2)$, proving (2.6). More explicitly, the preceding estimates give constants $C, \varepsilon_0 > 0$ such that, for $0 < |\varepsilon| < \varepsilon_0$,

$$\left| \mathcal{J}_{k,p}(\varepsilon) - \frac{2}{kp+2} - \frac{\varepsilon^2}{2}\mathcal{Q}_{k,p}(\lambda) \right| \le C\varepsilon^2 \omega_{kp}(|\varepsilon|), \tag{2.10}$$

where, for $0 < t < 1$,

$$\omega_\alpha(t) = \begin{cases} t^\alpha, & 0 < \alpha < 1, \\ t\log(e/t), & \alpha = 1, \\ t, & \alpha > 1. \end{cases}$$

For every fixed $\alpha > 0$, one has $\omega_\alpha(t) \to 0$ as $t \downarrow 0$. The expansion requires no uniform estimate as $p$ or $k$ varies. ∎

## 2.3 Proof of the counterexamples for every $0 < p < 1$

**Proof of Theorem 1.1.** Fix $0 < p < 1$ and an integer $k$ satisfying (1.1). The quadratic expression in (2.7) is minimized at

$$\lambda_{k,p} = \frac{(2-p)(kp+4)}{p(kp+2)} > 0. \tag{2.11}$$

Its minimum is

$$\begin{aligned} \mathcal{Q}_{k,p}(\lambda_{k,p}) &= \frac{p}{k} - \frac{(2-p)^2(kp+4)}{(kp+2)^2} \\ &= \frac{4\big((2-p)^2 - (1-p)(kp+2)^2\big)}{kp(kp+2)^2} < 0. \end{aligned} \tag{2.12}$$

Set $\gamma = -\mathcal{Q}_{k,p}(\lambda_{k,p}) > 0$. Apply (2.10) with $\lambda = \lambda_{k,p}$, and fix a sufficiently small $\varepsilon > 0$ such that $C\omega_{kp}(\varepsilon) \le \gamma/4$. The corresponding polynomial then satisfies

$$0 < A_\varepsilon := \int_{\mathbb{D}} |F_\varepsilon|^p \, dA \le \frac{2\pi}{kp+2} - \frac{\pi\gamma}{4}\varepsilon^2 < \frac{2\pi}{kp+2}. \tag{2.13}$$

Positivity follows because the coefficient of $z^k$ is 1. The choices of $\lambda$, $\varepsilon$, and the number $A_\varepsilon$ are now fixed in terms of $k, p$, before any $\xi$ is specified.

Let $\xi \in S_{H_k}$ have the form (2.3). For $\theta \in \mathbb{R}$, rotate the polynomial by

$$F_{\varepsilon,\theta}(z) := e^{-ik\theta} F_\varepsilon(e^{i\theta} z) = z^k + \varepsilon e^{-i\theta} z^{k-1} + \varepsilon \lambda_{k,p} e^{i\theta} z^{k+1}.$$

Rotation invariance gives $\int_{\mathbb{D}} |F_{\varepsilon,\theta}|^p \, dA = A_\varepsilon$. Since all coefficients below degree $k-1$ vanish, and $\xi$ has no entries above degree $k$,

$$(\xi \cdot F_{\varepsilon,\theta})(0) = 1 + \varepsilon c_{k-1} e^{-i\theta}.$$

Choose $\theta$ with $c_{k-1}e^{-i\theta} = |c_{k-1}|$; if $c_{k-1} = 0$, take any $\theta$. Then the normalized polynomial

$$G_\xi(z) = \frac{F_{\varepsilon,\theta}(z)}{1 + \varepsilon|c_{k-1}|}$$

satisfies $(\xi \cdot G_\xi)(0) = 1$ and

$$\int_{\mathbb{D}} |G_\xi|^p \, dA = \frac{A_\varepsilon}{(1+\varepsilon|c_{k-1}|)^p} \le A_\varepsilon.$$

Consequently, for every $\xi \in S_{H_k}$,

$$K_{\xi,\mathbb{D},p}(0) \ge \frac{1}{A_\varepsilon} > \frac{kp+2}{2\pi} = K_{\mathbb{D}}^{H_k,p}(0). \tag{2.14}$$

In particular, the positive number

$$\Delta_{k,p} := \frac{1}{A_\varepsilon} - \frac{kp+2}{2\pi} > 0$$

is independent of $\xi$. Thus (2.14) gives

$$\inf_{\xi \in S_{H_k}} K_{\xi,\mathbb{D},p}(0) \geq \frac{kp+2}{2\pi} + \Delta_{k,p} > K_{\mathbb{D}}^{H_k,p}(0). \tag{2.15}$$

Only the rotation angle and normalizing denominator depend on $\xi$; the fixed positive gap survives the infimum over the entire affine family. This proves (1.2).

Finally, every fixed $0 < p < 1$ admits an integer $k$ with

$$k > \frac{(2-p)/\sqrt{1-p} - 2}{p}, \tag{2.16}$$

and this is exactly (1.1). Thus the construction covers every exponent in $(0, 1)$. ■

## 2.4 The order and the first-order range

It is useful to describe precisely the range supplied by this second variation. For $k \geq 1$, set

$$p_k = \frac{k^2 - 4k - 1 + (k+1)\sqrt{k^2+6k+1}}{2k^2}. \tag{2.17}$$

For $0 < p < 1$, expansion of (1.1) gives

$$(1-p)(kp+2)^2 - (2-p)^2 = -p\big(k^2p^2 - (k^2-4k-1)p - 4k\big).$$

The polynomial in parentheses has one negative and one positive root; its positive root is $p_k$. Since its value at $p = 1$ is 1, one has $0 < p_k < 1$, and (1.1) is equivalent to $0 < p < p_k$. Moreover, for each fixed $p \in (0, 1)$ the left-hand side of (1.1) increases strictly with $k$ and tends to $+\infty$. Hence $p_k$ increases strictly to 1. For example,

$$p_1 = 2(\sqrt{2} - 1), \qquad p_2 = \frac{3\sqrt{17} - 5}{8}.$$

The values $p_k$ are the endpoints of the failure intervals proved by the present perturbation. No assertion is made that they are the exact global thresholds for fixed $k$.

Corollary 2.3. *Let $p_* = 2(\sqrt{2}-1)$. For every $0 < p < p_*$, the first-order model $H(z) = z$ satisfies*

$$\inf_{\xi \in S_H} K_{\xi,\mathbb{D},p}(0) > K_{\mathbb{D}}^{H,p}(0) = \frac{p+2}{2\pi}.$$

*In particular, at $p = 1/2$ the right-hand side is $5/(4\pi)$.*

**Proof.** Take $k = 1$ in Theorem 1.1. In this case

$$(1-p)(p+2)^2 - (2-p)^2 = -p(p^2+4p-4) > 0$$

exactly when $0 < p < 2(\sqrt{2}-1)$. ■

Remark 2.4. *For the first-order model, the minimum of the second variation is*

$$\mathcal{Q}_{1,p}(\lambda_{1,p}) = \frac{4(p^2+4p-4)}{(p+2)^2}.$$

*Its change of sign at $p_*$ concerns this fixed model. Theorem 1.1 reaches all $0 < p < 1$ by allowing the order to increase. Together with [3, Theorem 1.7], it shows that $p \geq 1$ is the optimal range for a $\xi$-reduction theorem valid for all orders.*

## 3 The $L^p$ analogue of $B = C$

Bao–Guan [2] proved that for $p = 2$,

$$B(F, I, D) := \sup_{\xi \in \ell_I} \frac{|(\xi \cdot F)(0)|^2}{K_{\xi,D,2}(0)} = C_{F,I}(D) := \inf\Big\{ \int_D |\widetilde{F}|^2 : (\widetilde{F} - F, 0) \in I,\ \widetilde{F} \in \mathcal{O}(D) \Big\},$$

where

$$\ell_I := \big\{ \xi \in \ell_1^{(n)} \setminus \{0\} : (\xi \cdot f)(0) = 0 \text{ for all } (f, 0) \in I \big\}.$$

For $p > 0$, define the corresponding quantities

$$B_p(F, I, D) := \sup_{\xi \in \ell_I} \frac{|(\xi \cdot F)(0)|^p}{K_{\xi,D,p}(0)},$$

$$C_{F,I,p}(D) := \inf\Big\{ \int_D |\widetilde{F}|^p : (\widetilde{F} - F, 0) \in I,\ \widetilde{F} \in \mathcal{O}(D) \Big\}.$$

The disk construction above disproves $B_p = C_{F,I,p}$ for every $0 < p < 1$. For $p \geq 1$, the identity holds for every ideal, as stated in Theorem 3.2 below.

THEOREM 3.1. *Let $0 < p < 1$, and let $k \geq 1$ satisfy (1.1). For $D = \mathbb{D}$, $F(z) = z^k$, and $I = \mathfrak{m}_0^{k+1}$,*

$$B_p(z^k, \mathfrak{m}_0^{k+1}, \mathbb{D}) < C_{z^k, \mathfrak{m}_0^{k+1}, p}(\mathbb{D}) = \frac{2\pi}{kp + 2}. \tag{3.18}$$

*Consequently, a counterexample exists for every $p \in (0, 1)$.*

**Proof.** The ideal condition is equivalent to $\widetilde{F}_0 = \cdots = \widetilde{F}_{k-1} = 0$ and $\widetilde{F}_k = 1$. Lemma 2.1 gives

$$C_{z^k, \mathfrak{m}_0^{k+1}, p}(\mathbb{D}) = \frac{2\pi}{kp + 2}.$$

The annihilator of $\mathfrak{m}_0^{k+1}$ consists of the nonzero functionals supported in degrees at most $k$:

$$\ell_I = \big\{ (\xi_0, \ldots, \xi_k, 0, \ldots) : (\xi_0, \ldots, \xi_k) \neq 0 \big\}.$$

If $\xi_k = 0$, then $(\xi \cdot z^k)(0) = 0$, so its contribution to $B_p$ is zero. If $\xi_k \neq 0$, divide $\xi$ by $\xi_k$ to obtain $\eta \in S_{H_k}$. Homogeneity gives

$$\frac{|(\xi \cdot z^k)(0)|^p}{K_{\xi,\mathbb{D},p}(0)} = \frac{1}{K_{\eta,\mathbb{D},p}(0)}.$$

All these kernels are positive and finite, since nonzero finite-jet functionals on $A^p(\mathbb{D})$ are bounded and act nontrivially on polynomials. Therefore

$$B_p(z^k, \mathfrak{m}_0^{k+1}, \mathbb{D}) = \sup_{\eta \in S_{H_k}} \frac{1}{K_{\eta,\mathbb{D},p}(0)} = \frac{1}{\inf_{\eta \in S_{H_k}} K_{\eta,\mathbb{D},p}(0)}.$$

The uniform bound (2.14) yields

$$B_p(z^k, \mathfrak{m}_0^{k+1}, \mathbb{D}) \leq A_\varepsilon < \frac{2\pi}{kp + 2},$$

which proves (3.18). ∎

THEOREM 3.2. *Let $p \geq 1$, let $D \subset \mathbb{C}^n$ be a pseudoconvex domain containing $o$, let $I$ be an ideal of $\mathcal{O}_o$, and let $(F, o) \in \mathcal{O}_o$. Then*

$$B_p(F, I, D) = C_{F,I,p}(D).$$

*Here both sides are allowed to be $+\infty$, with the conventions $a/0 = +\infty$ for $a > 0$, $0/0 = 0$, and $\sup \varnothing = 0$ in the definition of $B_p$.*

**Proof.** The inequality $B_p \leq C_{F,I,p}$ is elementary: for every admissible $\widetilde{F}$ and every $\xi \in \ell_I$ one has $(\xi \cdot F)(o) = (\xi \cdot \widetilde{F})(o)$, hence $|(\xi \cdot F)(o)|^p \leq K_{\xi,D,p}(o) \int_D |\widetilde{F}|^p$.

Assume first that $D$ is bounded. For the reverse inequality fix $k \geq 1$ and write $V_{I+\mathfrak{m}^k} := \{f \in A^p(D) : (f, o) \in I + \mathfrak{m}^k\}$. Since $\mathfrak{m}^k \subseteq I + \mathfrak{m}^k$, membership in $I + \mathfrak{m}^k$ is detected by the $(k-1)$-jet, so the quotient $Q_k := A^p(D)/V_{I+\mathfrak{m}^k}$ is finite-dimensional. As $p \geq 1$, $A^p(D)$ is a normed space, and $C_{F,I+\mathfrak{m}^k,p}(D)^{1/p} = \|[F]\|_{Q_k}$ (the class of $F$ being represented by the polynomial $j_{k-1}F$). By the Hahn–Banach theorem,

$$C_{F,I+\mathfrak{m}^k,p}(D)^{1/p} = \|[F]\|_{Q_k} = \sup\Big\{|\varphi(F)| : \ \varphi \in V_{I+\mathfrak{m}^k}^{\perp}, \ \|\varphi\| \leq 1\Big\}.$$

Every $\varphi \in V_{I+\mathfrak{m}^k}^{\perp}$ depends only on the $(k-1)$-jet, i.e. $\varphi(f) = (\xi \cdot f)(o)$ for some $\xi \in \ell_{I+\mathfrak{m}^k}$, and $\|\varphi\| = \|\xi\|_*$ with $\|\xi\|_* := K_{\xi,D,p}(o)^{1/p}$. Therefore

$$B_p(F, I + \mathfrak{m}^k, D) = C_{F,I+\mathfrak{m}^k,p}(D).$$

Since $I \subseteq I + \mathfrak{m}^k$, one has $\ell_{I+\mathfrak{m}^k} \subseteq \ell_I$, hence

$$B_p(F, I, D) \ \geq \ B_p(F, I + \mathfrak{m}^k, D) = C_{F,I+\mathfrak{m}^k,p}(D), \qquad k \geq 1.$$

It remains to note that $\lim_{k\to\infty} C_{F,I+\mathfrak{m}^k,p}(D) = C_{F,I,p}(D)$. Indeed $C_k := C_{F,I+\mathfrak{m}^k,p}(D)$ is increasing in $k$ with $C_k \leq C_{F,I,p}(D)$; if $C_\infty := \lim_k C_k < +\infty$, choose $G_k \in A^p(D)$ with $(G_k - F, o) \in I + \mathfrak{m}^k$ and $\int_D |G_k|^p \leq C_k + 1/k$. By Montel's theorem (through the submean-value property of $|f|^p$) a subsequence converges locally uniformly to some $G \in \mathcal{O}(D)$; Fatou's lemma gives $\int_D |G|^p \leq C_\infty$; and the jet conditions pass to the limit, so $(G - F, o) \in I + \mathfrak{m}^k$ for every $k$, whence $(G - F, o) \in I$ by the Krull intersection theorem [1, Chapter 10], $\bigcap_{k\geq 1}(I + \mathfrak{m}^k) = I$. Therefore $C_{F,I,p}(D) \leq C_\infty$, and combined with $C_\infty \leq C_{F,I,p}(D)$ this gives the limit. Consequently

$$B_p(F, I, D) \ \geq \ \lim_{k\to\infty} C_{F,I+\mathfrak{m}^k,p}(D) = C_{F,I,p}(D) \ \geq \ B_p(F, I, D),$$

which proves the theorem when $D$ is bounded.

For a general pseudoconvex domain, choose a bounded pseudoconvex exhaustion

$$D_1 \Subset D_2 \Subset \cdots \Subset D, \qquad o \in D_1, \qquad \bigcup_{j\geq 1} D_j = D.$$

The same Montel–Fatou argument used above gives

$$C_{F,I,p}(D_j) \nearrow C_{F,I,p}(D). \tag{3.19}$$

Indeed, the left-hand side is increasing and bounded above by the right-hand side; if its limit is finite, a diagonal limit of almost minimizers on $D_j$ is an admissible function on $D$ with no larger $L^p$ integral.

For a fixed $\xi \in \ell_I$, apply the same argument to

$$m_{\xi,p}(D_j) := \inf\Big\{\int_{D_j} |f|^p : f \in A^p(D_j),\ (\xi \cdot f)(o) = 1\Big\} = \frac{1}{K_{\xi,D_j,p}(o)}.$$

It follows that

$$m_{\xi,p}(D_j) \nearrow m_{\xi,p}(D), \qquad K_{\xi,D_j,p}(o) \searrow K_{\xi,D,p}(o), \tag{3.20}$$

where the extended values are permitted. Hence, using the monotonicity in $j$ to interchange the two suprema,

$$\begin{aligned} B_p(F,I,D) &= \sup_{\xi\in\ell_I} \lim_{j\to\infty} \frac{|(\xi\cdot F)(o)|^p}{K_{\xi,D_j,p}(o)} \\ &= \lim_{j\to\infty} \sup_{\xi\in\ell_I} \frac{|(\xi\cdot F)(o)|^p}{K_{\xi,D_j,p}(o)} = \lim_{j\to\infty} B_p(F,I,D_j) \\ &= \lim_{j\to\infty} C_{F,I,p}(D_j) = C_{F,I,p}(D), \end{aligned}$$

where the bounded case was used on every $D_j$. This completes the proof. ∎

Remark 3.3. *For $p = 2$, Theorem 3.2 is the result of [2]; the argument above is the same chain with the $L^2$ orthogonal projection replaced by the Hahn–Banach duality. For $F = z^k$ and $I = \mathfrak{m}_0^{k+1}$ on $\mathbb{D}$, the equality $B_p(F,I,\mathbb{D}) = C_{F,I,p}(\mathbb{D})$ is equivalent to the $\xi$-reduction formula for $H_k = z^k/k!$, by the reciprocal identities in the proof of Theorem 3.1. Theorems 1.1 and 3.1 show that both universal identities fail for every $0 < p < 1$, whereas Theorem 3.2 gives the ideal identity for all $p \geq 1$. Thus $p \geq 1$ is the optimal universal exponent range for both statements.*

## AI declaration

The first-order disk counterexamples in the original version were obtained with DeepSeek V4 Flash and Pro. The higher-order construction covering every $0 < p < 1$, and the corresponding proof revisions, were developed with assistance from Codex.

## References

[1] M. F. Atiyah and I. G. Macdonald. Introduction to Commutative Algebra. Addison–Wesley, Reading, MA, 1969.

[2] S. J. Bao and Q. A. Guan. Equivalence of the sharp effectiveness results of strong openness property. arXiv:2408.16372, 2024.

[3] S. J. Bao, Q. A. Guan and X. Sun, On the $p$-Bergman kernel with respect to a functional $\xi$, arXiv:2510.13144, 2025.